

Housing over the Life Cycle and Fertility

An illustrative model

Discussion draft for Tommaso De Santo September 8, 2026

Young households may have the lifetime resources to buy larger homes but lack the funds needed when they are young. We study when borrowing limits then keep housing with old owners. The welfare comparison allows households to adjust consumption and housing over their lives. Fertility is held fixed for this comparison; its response to housing finance is studied separately.

1 Environment

Households. At date t there are Y_t young and O_t old households. A young household has liquid wealth $b_i > 0$, income $y_i^y > 0$ when young, and income $y_i^o \geq 0$ when old. The triple (y_i^y, b_i, y_i^o) has distribution F in each entering cohort. Future income is known. Households choose completed fertility once and retain their tenure when old.

Preferences. Each child uses $\chi > 0$ units of goods and $\kappa > 0$ units of space. Total nondurable expenditure is c and housing is h ; the bundles left for adults are:

$$x = c - \chi n > 0, \quad s = h - \kappa n > 0. \quad (1)$$

Young and old utility are:

$$u_t^y(c, h, n) = \log(c - \chi n) + \alpha \log(h - \kappa n) + \vartheta_t \log n, \quad (2)$$

$$u^o(c^2, h^2, e) = \log c^2 + \gamma \log h^2 + \omega_B \log e. \quad (3)$$

All preference weights are positive. Households discount future utility by $\beta > 0$. Fertility $n > 0$ is continuous; the superscript 2 labels old-age quantities. The estate e consists of financial assets and the proceeds from selling housing at death. It enters the parent's utility but does not determine an entering household's wealth b_i .

A household draws an ownership preference ξ_i before making its choices. The draw is independent of its endowments and logistic with location $\bar{\xi}$ and scale $\sigma_\xi > 0$.

Housing and financial markets. The housing stock is $\bar{H} > 0$. Rental units satisfy $h \leq h_R^{\max}$ and owner units satisfy $h \leq h_O^{\max}$, where $0 < h_R^{\max} < h_O^{\max}$. Goods and bonds trade with the rest of the world. The bond price is $q \in (0, 1)$ and its gross return is $R_f = 1/q$. House prices are $P_t > 0$. Owners pay property tax at rate τ^p . Competition among rental intermediaries gives:

$$u_t \equiv qr_t = (1 + q\tau^p)P_t - qP_{t+1} > 0. \quad (4)$$

Here r_t is rent paid at the end of the period, and u_t is its value at the beginning. Property-tax revenue is rebated equally to young and old households; T_t denotes the rebate at the beginning of the period.

Home finance. Current income and liquid wealth are available at purchase. A mortgage finances at most the share $\phi_t \in (0, 1)$ of the house price. Principal and accumulated interest are repaid on entering old age. Unsecured borrowing is unavailable. Old owners can buy or sell housing within the owner size limit, using their income and wealth without new borrowing. Thus an old owner's choice is not bounded by the size of its previous home.

Demography. A child produces $\nu > 0$ young households next period, after survival and household formation. Cohort masses satisfy:

$$Y_{t+1} = \nu \bar{n}_t Y_t, \quad O_{t+1} = Y_t, \quad (5)$$

where $\bar{n}_t$ is average fertility among the young. Population here counts adult households.

2 Household choices

Young renters. Financial wealth on entering old age is a' . Conditional on renting, the young household solves:

$$W_t^R(i) = \max_{c,h,n,a'} u_t^y(c, h, n) + \beta V_{t+1}^R(a'; i),$$

$$c + qa' + u_t h = y_i^y + b_i + T_t, \quad a' \geq 0, \quad h \leq h_R^{\max}. \quad (6)$$

The renter pays for current housing services and saves for old age.

Young owners. For an owner, a' is financial wealth net of mortgage repayment. Its problem is:

$$W_t^O(i) = \max_{c,h,n,a'} u_t^y(c, h, n) + \beta V_{t+1}^O(a', h; i),$$

$$c + qa' + (1 + q\tau^p)P_t h = y_i^y + b_i + T_t,$$

$$qa' + \phi_t P_t h \geq 0, \quad h \leq h_O^{\max}. \quad (7)$$

To see the mortgage directly, let d be principal borrowed at purchase and k the amount invested in bonds. Then $0 \leq d \leq \phi_t P_t h$, $k \geq 0$, $a' = (k - d)/q$, and the budget is $c + k + (1 + q\tau^p)P_t h = y_i^y + b_i + T_t + d$. Eliminating k, d gives (7).

Old households. An old owner has net financial wealth a and title to H units of housing. Let $a^e \geq 0$ be financial saving during old age. The estate is:

$$e = \begin{cases} q^{-1}a^e, & \text{renter,} \\ q^{-1}a^e + P_{t+1}h^2, & \text{owner.} \end{cases} \quad (8)$$

The retained house is sold at death, at the end of old age. The two old-age problems are:

$$V_t^R(a; i) = \max_{c^2, h^2, e} u^o(c^2, h^2, e),$$

$$c^2 + qe + u_t h^2 = a + y_i^o + T_t, \quad h^2 \leq h_R^{\max}, \quad (9)$$

$$V_t^O(a, H; i) = \max_{c^2, h^2, e} u^o(c^2, h^2, e),$$

$$c^2 + qe + u_t h^2 = a + P_t H + y_i^o + T_t,$$

$$h^2 \leq h_O^{\max}, \quad e \geq P_{t+1}h^2. \quad (10)$$

The owner's estate restriction is equivalent to $a^e \geq 0$. Its cash budget before substituting for the estate is $c^2 + a^e + (1 + q\tau^p)P_t h^2 = a + P_t H + y_i^o + T_t$. Income, liquid wealth, and sale proceeds therefore enter the same budget.

Tenure. The household owns if $W_t^O(i) + \xi_i \geq W_t^R(i)$. Its ownership probability is:

$$\pi_t^O(i) = \frac{\exp\{[W_t^O(i) + \bar{\xi}]/\sigma_\xi\}}{\exp\{W_t^R(i)/\sigma_\xi\} + \exp\{[W_t^O(i) + \bar{\xi}]/\sigma_\xi\}}. \quad (11)$$

The taste draw affects tenure but not choices conditional on tenure.

3 Equilibrium and financing

An equilibrium determines house prices, rebates, household choices, and the distribution of assets and housing across cohorts. Let G_t denote the distribution of old households over financial wealth, housing titles, old income, and tenure. Housing and the government budget satisfy:

$$Y_t \bar{h}_t^y + O_t \bar{h}_t^o = \bar{H}, \quad (Y_t + O_t)T_t = q\tau^p P_t \bar{H}. \quad (12)$$

The bars average occupied housing over types and tenure choices.

Definition 1. *Given the initial cohorts and old distribution, an equilibrium is a sequence of prices, rebates, household choices, tenure probabilities, and cohort distributions such that households solve (6)–(10), tenure follows (11), rental entry satisfies (4), markets satisfy (12), and cohorts evolve according to (5) and the young households' choices.*

At a positive stationary equilibrium, prices, distributions, and cohort masses are constant. In particular:

$$Y = O = N, \quad \bar{n} = 1/\nu, \quad N = \frac{\bar{H}}{\mathbb{E}[h^y + h^o]}. \quad (13)$$

The housing stock determines cohort mass once stationary household choices are known. Appendix A gives explicit conditions for existence with heterogeneous endowments and positive property taxes.

Current and lifetime resources. At stationary prices write $w = y_i^y + b_i + T$, $v = y_i^o + T$, and:

$$p = (1 - q + q\tau^p)P, \quad L = (1 - \phi + q\tau^p)P. \quad (14)$$

Here p is the service cost of housing and L is its cash requirement. For an owner with old-age resources $z = a' + Ph + v$, the two budgets become:

$$c + ph + qz = w + qv, \quad c + Lh \leq w. \quad (15)$$

The first values the household's lifetime resources; the second requires current consumption and the cash portion of housing to be paid when young.

When old housing is below its cap, the old value has the form $(1 + \gamma + \omega_B) \log z + \text{constant}$, even if its estate floor binds. Write $K = 1 + \gamma + \omega_B$, $B = \beta K$, and $D = 1 + \alpha + \vartheta + B$. If the young and old caps are slack at the allocation without the mortgage restriction, that allocation has:

$$\begin{aligned} c^* &= \frac{w + qv}{D} \left(1 + \frac{\vartheta\chi}{\chi + p\kappa} \right), \\ h^* &= \frac{w + qv}{D} \left(\frac{\alpha}{p} + \frac{\vartheta\kappa}{\chi + p\kappa} \right). \end{aligned} \quad (16)$$

The mortgage limit strictly restricts the household precisely when $c^* + Lh^* > w$. Equivalently:

$$\boxed{\frac{w + qv}{w} > \frac{D}{1 + \alpha L/p + \vartheta(\chi + L\kappa)/(\chi + p\kappa)}}. \quad (17)$$

The household wants an allocation whose current consumption and down payment exceed current resources. If a physical cap binds, the same cash test applies to the allocation computed with that cap retained.

4 Housing allocation and welfare

We ask whether financing limits prevent a mutually beneficial reallocation of housing over the life cycle. The planner may relax these financing limits and make household-specific transfers and loans with future repayment. It respects the housing stock, tenure size limits, existing creditors' claims, and the external bond return. Fertility and tenure are held fixed. Selected households may change their own old-age consumption, housing, and estates; other households must be compensated.

Proposition 1 (Housing over the life cycle). *Consider a strictly mortgage-constrained young owner whose young and old housing caps and old estate floor are slack. If*

$$\boxed{\beta(1 + \gamma + \omega_B)(1 - \phi + q\tau^P) \geq \phi - q,} \quad (18)$$

relaxing its financing restriction, at fixed prices and fixed fertility, increases its desired young housing and reduces its old-age housing. A finite step in this direction is feasible below the physical owner cap and strictly improves lifetime utility.

The condition permits $q < \phi$. Greater weight on old-age utility makes it easier to satisfy. This result concerns the household's allocation over its life: young housing need not initially have a higher marginal value in units of current consumption than old housing.

Proof. Hold equilibrium fertility n fixed. Set $w_n = w - (\chi + L\kappa)n$ and $M_n = w + qv - (\chi + p\kappa)n$. The adult allocation solves:

$$\max \log x + \alpha \log s + B \log S, \quad x + ps + S = M_n, \quad x + Ls \leq w_n, \quad (19)$$

where $S = qz$. Without the financial restriction its choice is $(x_F, s_F, S_F) = M_n(1, \alpha/p, B)/(1 + \alpha + B)$.

Let $\lambda, \mu > 0$ be the lifetime-budget and cash multipliers at the constrained allocation. Define $r = \mu/(\lambda + \mu)$, $\rho = (1 - r)p + rL$, and $j = 1/(\lambda + \mu)$. The first-order conditions give $(x_C, s_C, S_C) = (j, \alpha j/\rho, Bj/(1 - r))$. Subtraction yields:

$$s_C - s_F = \frac{\alpha jr\{(p - L)(1 - r) - BL\}}{p(1 + \alpha + B)\rho(1 - r)}. \quad (20)$$

Condition (18) is $BL \geq p - L$ and makes this expression strictly negative. Also:

$$\frac{S_C}{M_n} = \frac{B}{B + (1 - r)(1 + \alpha p/\rho)} > \frac{B}{1 + \alpha + B} = \frac{S_F}{M_n}. \quad (21)$$

Old resources, and therefore old housing, fall. A sufficiently small convex step toward the unrestricted allocation preserves the young cap, the old cap, and the estate floor while strictly increasing utility. $\square$

The inequality is sufficient rather than necessary. When $p > L$, an exact condition for a strictly constrained household to prefer larger young housing in the fixed-fertility comparison is:

$$Lqv > (p - L)(w - \chi n). \quad (22)$$

Enough future income relative to current resources also delivers the direction when (18) fails.

Proposition 2 (Inefficiency). *Suppose the stationary equilibrium has a positive share of young owners satisfying Proposition 1. It admits a Pareto improvement that transfers housing from old owners to young owners at the intervention date. Fertility, tenure, and every future cohort size are unchanged. New external borrowing is repaid by the selected households' estate date.*

The required owner share follows from explicit income, preference, and capacity inequalities in Appendix A; it is not an assumed difference in marginal housing values. Under those inequalities a stationary equilibrium exists, and every stationary equilibrium satisfies the proposition's conditions.

The argument uses two housing reallocations. Current old owners supply the additional homes needed by the selected young. When those young households become old, their smaller homes

release space to other old owners. The latter groups are compensated exactly. A small portion of the selected households' lifetime gain covers the compensation costs. Appendix B gives the transfers and loan repayment.

This establishes one Pareto improvement with the stated housing direction. It does not characterize the entire first-best allocation or establish that a uniform mortgage reform is Pareto improving.

5 Fertility and ownership

For comparisons at a given transition date, the formulas in this section use $p = u_t$ and $L = (1 - \phi_t + q\tau^p)P_t$, with $v = y_i^o + T_{t+1}$. The household chooses children by comparing their benefit with their goods and space costs:

$$\frac{\vartheta}{n} = \frac{\chi}{c - \chi n} + \frac{\alpha\kappa}{h - \kappa n}. \quad (23)$$

More housing lowers the space cost of another child at fixed consumption. Along a change in both c and h , differentiation gives:

$$dn = \frac{(\chi/x^2)dc + (\alpha\kappa/s^2)dh}{\vartheta/n^2 + \chi^2/x^2 + \alpha\kappa^2/s^2}. \quad (24)$$

The financing of additional housing therefore matters for fertility.

Proposition 3 (Fertility under easier mortgage finance). *Hold the price and rebate path fixed, and suppose old housing caps are slack at the relevant conditional choices. A strictly constrained owner increases fertility when ϕ rises if $L \geq p$. When $L < p$, a sufficient condition is:*

$$Bp \left(L + \frac{\chi}{\kappa} \right) \geq (p - L) \left(p - \alpha \frac{\chi}{\kappa} \right). \quad (25)$$

With slack young housing, this condition is sharp for a fertility increase uniform over current and future incomes at given L, p .

This restriction compares the goods and space costs of children and the weight on future utility. It contains no borrowing multiplier. The simpler condition $(\alpha + B)\chi \geq p\kappa$ guarantees the fertility direction throughout any range of nonnegative L . A binding young housing cap fixes housing; easier finance then raises fertility by releasing current resources.

Tenure changes require one additional comparison. A higher ϕ expands the owner's feasible set, so ownership weakly increases at fixed prices. For each type, mean fertility changes according to:

$$\frac{d\bar{n}_i}{d\phi} = \pi_i^O \frac{dn_i^O}{d\phi} + \frac{d\pi_i^O}{d\phi} (n_i^O - n_i^R). \quad (26)$$

The fertility difference between owners and renters determines the contribution of households switching tenure.

Proposition 4 (Fertility including tenure choice). *Suppose $L \leq p$, old housing caps are slack at both conditional choices, and:*

$$\frac{p}{\alpha + B} \leq \frac{\chi}{\kappa} \leq \frac{p}{\alpha}. \quad (27)$$

Then $n_i^O \geq n_i^R$, and easier mortgage finance weakly increases fertility averaged over a fixed heterogeneous cohort, including tenure changes. The result extends to finite reforms for which the conditions hold throughout.

Renting is the same current-resource problem with cash cost $L = p$ and the smaller rental cap. Under the upper bound in (27), access to larger housing at $L = p$ weakly raises fertility. Under the lower bound, lowering L to the owner's cash cost weakly raises it further. This gives the tenure ordering; both terms in (26) are then nonnegative. The proof and the sharp individual condition are given in Appendix C.

These are household and fixed-cohort results. The price p is endogenous in equilibrium. A mortgage reform can change prices and rebates as well as the household's financing opportunities; the displayed restrictions do not by themselves sign that general-equilibrium response.

6 A fertility decline and a later housing policy

The intended experiment starts from a stationary equilibrium. A persistent decline in ϑ_t initiates demographic adjustment. At a later date a housing policy is introduced, and its path is compared with continued adjustment without the policy. The two paths share the inherited state and non-policy shocks at announcement. If policy is anticipated, expectations must enter choices from that earlier date.

At fixed prices and rebates, a decline in ϑ weakly reduces fertility, including tenure selection. For each type, the envelope theorem gives $\partial(W^O - W^R)/\partial\vartheta = \log(n^O/n^R)$. Consequently:

$$\begin{aligned} \frac{\partial \bar{n}_i}{\partial \vartheta} &= \pi_i^O \frac{\partial n_i^O}{\partial \vartheta} + (1 - \pi_i^O) \frac{\partial n_i^R}{\partial \vartheta} \\ &\quad + \frac{\pi_i^O(1 - \pi_i^O)}{\sigma_\xi} (n_i^O - n_i^R) \log(n_i^O/n_i^R) \geq 0. \end{aligned} \quad (28)$$

Conditional fertility is nondecreasing in its preference weight, and the selection term is nonnegative regardless of which tenure has higher fertility.

For any two existing equilibrium paths with the same young cohort at date t_0 , demographic accounting implies:

$$\frac{Y_t^{\text{policy}}}{Y_t^{\text{baseline}}} = \prod_{s=t_0}^{t-1} \frac{\bar{n}_s^{\text{policy}}}{\bar{n}_s^{\text{baseline}}}. \quad (29)$$

This relates population differences to fertility differences accumulated along the path. At positive stationary endpoints, fertility returns to $1/\nu$ in both economies; their population levels can differ.

At those endpoints, the population comparison is:

$$\frac{Y^{\text{policy}} + O^{\text{policy}}}{Y^{\text{baseline}} + O^{\text{baseline}}} = \frac{\mathbb{E}[h^y + h^o]^{\text{baseline}}}{\mathbb{E}[h^y + h^o]^{\text{policy}}}. \quad (30)$$

With fixed total housing, a larger stationary population requires less housing per household over the life cycle, after prices and tenure have adjusted.

A transition existence or sign theorem requires more than stationary existence. At every date the housing caps require $Y_t + O_t \geq \bar{H}/h_O^{\max}$, and an inherited owner must have $a + P_t H + y_i^o + T_t > 0$ to finance positive old-age consumption and an estate. The current results do not establish an invariant region satisfying these requirements or convergence after a finite preference decline. Equations (29)–(30) are exact conditional comparisons, not a proof that a mortgage policy raises fertility at every transition date or the final population. The earlier model's transition construction cannot be transferred to this specification unchanged.

A Primitive conditions and stationary existence

This appendix supplies conditions on endowments and parameters that ensure the stationary equilibrium and owner groups used in the welfare proposition. They are sufficient and conservative. No price or borrowing multiplier is an input to these conditions.

Assume bounded endowments, $w_{0i} = y_i^y + b_i \geq \underline{w} > 0$ and $v_{0i} = y_i^o \geq 0$, and $0 \leq \tau^p < 2$. Define:

$$\begin{aligned} d_p &= 1 - q + q\tau^p, & d_L &= 1 - \phi + q\tau^p, & d_{\max} &= \max\{d_p, d_L\}, \\ \bar{M}_0 &= \mathbb{E}[w_0 + qv_0], & \bar{T} &= \frac{\tau^p \bar{M}_0}{(1 - q)(2 - \tau^p)}, & \bar{M} &= \bar{M}_0 + (1 + q)\bar{T}. \end{aligned} \quad (31)$$

The scalar $\bar{T}$ bounds a balanced stationary rebate. Suppose:

$$h_R^{\max} > \frac{\kappa}{\nu}, \quad \vartheta\nu > \frac{\chi D}{\underline{w}} + \frac{\alpha\kappa}{h_R^{\max} - \kappa/\nu}. \quad (32)$$

These conditions ensure that sufficiently inexpensive housing permits fertility above replacement. They do not require renters to have fertility above replacement at the equilibrium price.

At least one positive stationary equilibrium exists, and every stationary equilibrium satisfies:

$$P_- < P < P_+, \quad 0 \leq T \leq \bar{T}, \quad P_- \equiv \frac{\alpha\underline{w}}{Dd_{\max}h_R^{\max}}, \quad P_+ \equiv \frac{\nu\bar{M}}{d_p\kappa}. \quad (33)$$

To establish this, combine the two age budgets. In either tenure:

$$x + ps + (\chi + p\kappa)n + qc^2 + qph^2 + q^2e = w + qv. \quad (34)$$

Scaling the first-order conditions, including both caps and the homogeneous estate floor, gives $\lambda(w + qv) + \mu w \leq D$, while $1/x = \lambda + \mu$. Thus $x \geq w/D \geq \underline{w}/D$. If the young housing cap is slack, $\alpha x/s \leq d_{\max}P$. If it binds, housing is at least $h_R^{\max}$. At $P \leq P_-$ every conditional choice therefore has $h \geq h_R^{\max}$. The fertility condition (23) and (32) then rule out $n \leq 1/\nu$.

Conversely, (34) implies $\bar{n} < \{\bar{M}_0 + (1 + q)T\}/(\chi + d_p P\kappa)$, so fertility is below replacement for $P \geq P_+$ and $T \leq \bar{T}$. Also, if $\bar{h} = \mathbb{E}[h^y + h^o]$, the balanced-rebate map satisfies:

$$\mathcal{T}(P, T) = \frac{q\tau^p P \bar{h}(P, T)}{2} \leq \frac{\tau^p}{2d_p} \{\bar{M}_0 + (1 + q)T\} \leq \bar{T}. \quad (35)$$

The last inequality follows from $2d_p - \tau^p(1 + q) = (1 - q)(2 - \tau^p) > 0$.

Conditional allocations are unique and continuous, including where caps change status. Logistic tenure probabilities preserve aggregate continuity. Map T to $\mathcal{T}(P, T)$ and move P in the direction of $\nu\bar{n}(P, T) - 1$, restricting the new price to $[P_-, P_+]$. Brouwer's theorem gives a fixed point. The strict fertility signs exclude the price boundaries, so it has replacement fertility. Setting $N = \bar{H}/\bar{h}$ closes housing and the fiscal budget, and the young choices generate the stationary old distribution. The same bounds apply to any stationary equilibrium. Uniqueness and stability are separate questions.

Eligible young owners. Put $\ell = d_L/d_p$ and:

$$C_{\min} = 1 + \alpha\ell + \vartheta \min\{1, \ell\}, \quad k = \frac{D}{C_{\min}} - 1. \quad (36)$$

Choose a positive- F -mass set $\mathcal{S}$ of endowments satisfying:

$$qv_{0i} > kw_{0i} + \max\{k - q, 0\}\bar{T} \quad (i \in \mathcal{S}). \quad (37)$$

This ensures enough future income relative to current resources for finance to bind, even after the common rebate. Let $M_S \geq \sup_{i \in \mathcal{S}} \{w_{0i} + qv_{0i} + (1+q)\bar{T}\}$ and impose:

$$\omega_B d_p > q\gamma, \quad h_O^{\max} > \frac{M_S}{d_p P_-} \max \left\{ 1, \frac{\gamma}{qK} \right\}. \quad (38)$$

The first condition makes the old estate floor slack. The lifetime budget bounds young housing by M/p and old resources by M/q , so the second makes both caps slack in the actual and finance-relaxed choices of these types. This cap bound uses the renter cap in P_- ; enlarging the owner cap does not lower the price bound.

The bracket in the cash-demand test (17) is at least $C_{\min}$. Condition (37) implies $(w + qv)/w > D/C_{\min}$ throughout $T \in [0, \bar{T}]$. Thus finance is strictly restrictive for every type in $\mathcal{S}$ at every stationary equilibrium. The logistic distribution assigns each type a positive owner probability. Their predecessors supply current old owners, and unselected contemporaries supply next period's old recipients. There is no separate equal-mass assumption on young and old income groups.

These conditions, together with (18), imply Proposition 2. They establish an equilibrium result in primitives. Their conservative bounds should not be read as evidence that they fit a particular calibration.

B Compensation and external repayment

This appendix gives the construction first for common conditional bundles; the same bounds apply uniformly on positive-mass subsets of heterogeneous types. Hold every selected household's fertility and tenure fixed. Take a finite improving step from Proposition 1. Write its changes as $(\Delta c, \Delta h, \Delta c^2, \Delta h^2, \Delta e)$, with $\Delta h > 0$ and $\Delta h^2 < 0$. The lifetime budget gives:

$$\Delta c + p\Delta h + q\Delta c^2 + qp\Delta h^2 + q^2\Delta e = 0. \quad (39)$$

If the step gives utility gain $g > 0$ and new adult consumption $x_A > 0$, deduct $\eta = x_A(1 - e^{-g/2})$. This leaves gain $g/2$ and releases η goods per selected household.

Choose a small positive mass ϵ of such households. Current old donors, of mass $m_D > 0$, each surrender $s_D = \epsilon\Delta h/m_D$ housing. Hold their estates fixed and give each:

$$D(s_D) = c_D \{(1 - s_D/h_D)^{-\gamma} - 1\}. \quad (40)$$

This preserves old utility exactly. Next period give $s_R = \epsilon(-\Delta h^2)/m_R$ extra housing to each member of an unselected old owner group of mass $m_R > 0$. Keeping their estates fixed, collect:

$$R(s_R) = c_R \{1 - (1 + s_R/h_R)^{-\gamma}\}. \quad (41)$$

Their lifetime utilities are also unchanged. Small ϵ preserves their old size caps and positive financial estates. Housing clears at both dates, within the owner segment.

Both old groups have $\gamma c/h = p$ before the transfer. Their total compensation costs therefore equal $p\epsilon\Delta h$ at the first date and their total payments equal $p\epsilon(-\Delta h^2)$ at the next, up to terms bounded by $C\epsilon^2$ for a finite constant C . The net present goods requirement is at most $C\epsilon^2 - \epsilon\eta < 0$ for sufficiently small positive ϵ . This is an analytical bound from the second derivatives of the two displayed compensation functions.

In detail, the additional goods uses at the three affected dates are:

$$\begin{aligned} G_0 &= \epsilon(\Delta c - \eta) + m_D D(s_D), \\ G_1 &= \epsilon\Delta c^2 - m_R R(s_R), \\ G_2 &= \epsilon\Delta e. \end{aligned} \quad (42)$$

The terminal balance of new external borrowing is $G_0/q^2 + G_1/q + G_2 \leq 0$. Thus every new loan is repaid by the selected households' estate date. Internal housing title payments cancel. Existing mortgage claims and the other households' estates are preserved. Selected households may leave smaller estates as part of their preferred lifetime allocation; that utility cost is included in g .

For heterogeneous groups, first reserve a positive-mass group of future old recipients, and then select reform households from the remaining eligible owners. Restrict to positive-mass subsets with uniform strict margins and $\eta_i \geq \underline{\eta} > 0$, $\Delta h_i \leq \overline{H}$, $-\Delta h_i^2 \leq \overline{K}$. These subsets exist by taking countable level sets of the positive gains and margins. The continuous ownership taste allows any sufficiently small selected mass, even if the income distribution has atoms. Define total housing changes by:

$$H_\epsilon = \int_{S_\epsilon} \Delta h_i d\mu, \quad K_\epsilon = - \int_{S_\epsilon} \Delta h_i^2 d\mu. \quad (43)$$

Take H_ϵ/m_D from each current donor and give K_ϵ/m_R to each future recipient, applying that household's own compensation function. If $\overline{A}_D, \overline{A}_R$ bound their second derivatives, the cost above the linear housing value is at most:

$$\frac{\overline{A}_D H_\epsilon^2}{2m_D} + q \frac{\overline{A}_R K_\epsilon^2}{2m_R} \leq C\epsilon^2. \quad (44)$$

Fees yield at least $\underline{\eta}\epsilon$. Choose selected mass below $\underline{\eta}/(2C)$ and small enough to preserve the uniform donor and recipient margins. This establishes the same strict resource surplus with unequal group masses and heterogeneous individual bundles.

Individual budgets. Aggregate resources can be implemented while preserving each private mortgage. Write $A = (1 + q\tau^p)P = p + qP$. For selected household i , hold its original private net financial claim $a'_{C,i}$ fixed and give transfers:

$$f_{0i} = \Delta c_i - \eta_i + A\Delta h_i, \quad f_{1i} = \Delta z_i - P\Delta h_i, \quad f_{0i} + qf_{1i} = -\eta_i. \quad (45)$$

This is a loan of $q(P\Delta h_i - \Delta z_i) > 0$ when young, repaid on entering old age, together with fee η_i . After repaying, its old resources are exactly $z_{C,i} + \Delta z_i > 0$; its new old financial saving is $q(e_{A,i} - Ph_{A,i}^2) > 0$. No new borrowing in old age is needed.

A donor who gives up s_D receives the additional transfer $f_{Di} = D_i(s_D) - ps_D$ and raises old financial saving by qPs_D , keeping its estate fixed. A recipient who obtains s_R receives $f_{Ri} = ps_R - R_i(s_R)$ and reduces financial saving by qPs_R , within its strict margin. The program's present-value cost is:

$$\int_{S_\epsilon} (f_{0i} + qf_{1i}) d\mu + \int_D f_{Di} d\mu + q \int_R f_{Ri} d\mu \leq C\epsilon^2 - \underline{\eta}\epsilon < 0. \quad (46)$$

The program can settle its account at the next date; the selected old households' portfolios finish at their estate date. At the first date extra young titles equal current donors' title reductions. At the next date those inherited titles offset the donors' smaller death sales, and selected old housing reductions equal recipients' additions. Their title changes at death therefore cancel. Future entrants can make their original purchases, and all originally contracted creditors receive their promised payments.

C Fertility signs

Write $\eta = \chi/\kappa$, rescale children as $k = \kappa n$, and let $S = qz$. At a strictly constrained owner optimum with slack caps, put $t = Bx/S \in (0, 1)$ and $\rho = \alpha x/s = (1 - t)L + tp$. These quantities are used

in the proof, not in the statement of the fertility condition. Define $H(\rho) = \alpha/\rho + \vartheta/(\eta + \rho)$ and $A = \rho^2/\alpha$. Differentiating the two choice equations for h, k shows that the fertility derivative has the sign of:

$$\mathcal{N} = (1 - t)(A - L\eta) + H(\rho) \left[A(L + \eta) + \frac{t^2(L - p)}{B}(A - p\eta) \right]. \quad (47)$$

The Hessian determinant is positive, by strict concavity.

The expression excluding its last term is bounded below by $(1 - t)A + \rho L + \eta tp > 0$. For $L \geq p$, if the last term is negative, $S = qv + (L - p)h$ and $v \geq 0$ give $t(L - p)H(\rho) \leq B$. This bounds the entire expression below by $A + \rho L > 0$.

For $L < p$, the last term is nonnegative when $A \leq p\eta$. Otherwise the first term is positive and $t^2(1 - p\eta/A) < 1 - \alpha\eta/p$. Condition (25) makes the remaining bracket positive. As t approaches one from below, its limiting sign is the sign of:

$$\frac{p^2}{\alpha}(L + \eta) - \frac{p - L}{B} \left(\frac{p^2}{\alpha} - p\eta \right). \quad (48)$$

If the stated condition fails, this expression is negative, and nearby strictly constrained income profiles give decreasing fertility. Such profiles are feasible: choose any $x > 0$ and set $w = x\{1 + \alpha + \vartheta + t(L - p)H(\rho)\}$ and $qv = x\{B/t - (L - p)H(\rho)\}$.

If the young cap binds, differentiation at fixed housing gives:

$$\frac{\partial n}{\partial \phi} = \frac{P\chi h/x^2}{\vartheta/n^2 + \chi^2/x^2 + \alpha\kappa^2/s^2} > 0. \quad (49)$$

Continuity at changes of constraint status extends the weak sign to finite reforms that satisfy the condition throughout.

For the tenure comparison, first set $L = p$ and raise the young cap from $h_R^{\max}$ to $h_O^{\max}$. If finance binds, the sign of the fertility response is the sign of $\alpha\kappa/s^2 - \chi p/x^2$. The cap condition $\alpha x/s \geq p$ and $\chi/\kappa \leq p/\alpha$ make it nonnegative. If finance is slack, the same argument replaces χp by $\chi p/(1 + B)$ and is also nonnegative. Lowering L to its actual owner value then raises fertility under $(\alpha + B)\chi \geq p\kappa$. Old conditional-value constants affect tenure probabilities but not these conditional fertility comparisons.

D Repayment timing and the existing note

The proposal changes three features of the existing note: current income is available at purchase, households have an old-age income endowment, and old owners can resize their homes. It keeps the original mortgage repayment convention. The welfare comparison permits changes to selected households' own future allocations; this is essential for Proposition 1 when $q < \phi$.

For literature context, van Doornik, Fazio, Ramadorai, and Skrastins's *Housing and Fertility* (December 2024, Section 2) provides a simple goods-and-space account of fertility but does not model endogenous tenure choice or a mortgage down payment. Coven, Golder, Gupta, and Ndiaye's *Property Taxes and Housing Allocation Under Financial Constraints* (August 2026, Section 3.1) derives capitalization and intergenerational redistribution in a simple collateral model. Its old households sell housing; it does not supply the old-versus-young housing-services welfare result above. These are related ingredients, not proofs of this model's conclusions.